ACTUARIAL RESEARCH REPORT — VERSION 2

Ind AS 117 (IFRS 17) Life Insurance CSM Model

Indian Market • IALM 2012-14 • RBI G-Sec Discount Rates • Monte Carlo Risk Adjustment

Field	Detail
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IFoA Exemptions	CM1, CM2, CB1, CB2, CS1, CS2
Date	FY2025
Language	Python 3 (numpy • scipy • matplotlib • pandas)
Regulation	Ind AS 117 — equivalent to IFRS 17 (MCA notified August 2024)
IRDAI Deadline	FY27 (April 2027) — all Indian insurers
Report version	v2 — enhanced with full technical assumptions, policy-type analysis,
	age/gender sensitivity, Monte Carlo methodology, and claim timing

EXECUTIVE SUMMARY

Ind AS 117 (IFRS 17) Life Insurance CSM Model

Indian Market • IALM 2012-14 Mortality • RBI G-Sec Discount Rates • Monte Carlo Risk Adjustment • Python 3

What this project is

A fully working actuarial model implementing India's Ind AS 117 (equivalent to IFRS 17) for life insurance. It covers the complete General Measurement Model workflow — loading official IALM 2012-14 mortality tables, building an RBI G-Sec discount curve, calculating all three balance sheet components (BEL, RA, CSM), and producing the 20-year CSM roll-forward that governs how insurance profit flows into P&L. All data sources are real and India-specific. The Risk Adjustment is calculated via Monte Carlo simulation — 500 scenarios, each randomly drawing death or survival for every policyholder every year using np.random.binomial() — the methodologically correct approach that a spreadsheet cannot replicate.

Why it matters right now

IRDAI has mandated full Ind AS 117 implementation by April 2027. All 58 Indian insurers managing ₹8.30 lakh crore in premiums must transition from Indian GAAP to this framework. The first proforma submission cycle completed December 2025; the second is due June 2026. Indian actuaries are actively building this type of model right

Key results at a glance

Component	₹ crore	What it means
Best Estimate Liability (BEL)	−1.607	
Risk Adjustment (RA) — 75th pctile	+0.997	
Contractual Service Margin (CSM)	+2.345	
Total Insurance Contract Liability	+1.735	

Portfolio snapshot

Policies simulated	1,000 term life policies
Avg entry age / term	40.1 years / 20-year policy
Avg sum assured	₹27.7 lakh (log-normal)
Gender mix	70% male / 30% female
Avg gross premium	₹18,264 per policy p.a.

now. This project demonstrates the end-to-end capability required: survival probability calculations, EPV pricing, market-consistent discounting, stochastic simulation, and CSM accounting mechanics — all in a single reproducible Python codebase.

Tools and data

Item	Detail
Language	Python 3 — numpy, scipy, matplotlib, pandas
Mortality	IALM 2012-14 — Institute of Actuaries of India
Discount rates	RBI G-Sec — CCIL / RBI DBIE FY2025
Industry data	IRDAI Annual Report 2024-25
Portfolio	1,000 policies • 500 Monte Carlo scenarios

Three things to take away

1 Negative BEL is not a mistake. It means the insurer expects to collect more in premiums than it pays in claims and expenses. That surplus becomes the CSM — it never hits P&L on day one.

2 The CSM of ₹2.345 cr is deferred profit. It sits on the liability side of the balance sheet and releases to P&L gradually over 20 years — ₹0.131 cr in year 1, rising to ₹0.389 cr in year 20. It reaches exactly zero at maturity.

3 The ₹0.997 cr Risk Adjustment comes from real simulation. 500 Monte Carlo scenarios each randomly drew death or survival per policyholder per year using np.random.binomial(). The 75th percentile minus the mean is what the insurer charges for bearing uncertainty.

Where to find everything

GitHub	https://github.com/prithiyangabintumani/IndAS117-CSM-Model
LinkedIn	https://linkedin.com/in/prithiyangabintumani
Full report	`Sections 3-5: methodology, assumptions, interpretation`

1. Project Overview

This report documents an end-to-end actuarial model built under Ind AS 117 — India's adoption of the International Financial Reporting Standard 17 (IFRS 17) for insurance contracts. The model implements the General Measurement Model (GMM), also called the Building Block Approach, across a simulated portfolio of 1,000 Indian term life insurance policies. This second version of the report significantly expands the technical treatment of key assumptions, including age and gender distributions, policy type scope, claim payment timing, Monte Carlo methodology, and the factors that drive the Contractual Service Margin.

India's Insurance Regulatory and Development Authority (IRDAI) has mandated Ind AS 117 implementation by April 2027 (FY27). This deadline affects all 58 registered Indian insurers — 25 life and 33 general — representing a combined premium base of ₹8.30 lakh crore (IRDAI Annual Report 2024-25). The transition requires insurers to shift from Indian GAAP to a fulfilment-value, market-consistent basis of financial reporting, fundamentally changing how insurance profit is recognised and when losses must be disclosed.

1.1 Regulatory Context

Milestone	Date	Detail
ICAI Exposure Draft	2018	Initial Ind AS 117 proposed to align with IFRS 17
IRDAI selects 15 firms for early adoption	August 2023	Foreign equity partner insurers required to adopt
IRDAI Expert Committee reconstituted	February 2024	Phase-wise implementation roadmap established
MCA notifies Ind AS 117	August 2024	Official Gazette notification G.S.R. 492(E)
First proforma submission	December 2025	Gap assessment cycle 1 — all insurers
Second proforma submission	June 2026	Gap assessment cycle 2
Full implementation deadline	April 2027	All Indian insurers — FY27 onwards

1.2 Why Indian GAAP Was Replaced

The move from Indian GAAP to Ind AS 117 addresses four structural problems that made Indian insurance financial statements unreliable for investors, regulators, and policyholders:

- **Inconsistency across insurers:** Different companies used different reserving methods, assumption bases, and profit recognition policies. There was no common framework for comparing LIC, HDFC Life, SBI Life, or any other insurer on a like-for-like basis.
- **Day-one profit front-loading:** Under Indian GAAP, a significant portion of the profit embedded in a new long-term policy could be recognised in the first year on receipt of the initial premium — before the insurance risk was actually borne. This created incentives for volume-driven selling with aggressive pricing assumptions. The Contractual Service Margin eliminates this entirely.

- **Stale liability measurement:** Indian GAAP reserves were set using assumptions locked in at policy inception and not updated for current market conditions. If interest rates fell after policies were sold, the true economic liability was understated. Ind AS 117 requires current market discount rates for the BEL, creating a live economic picture.
- **Barriers to global capital:** India's 'Insurance for All by 2047' target requires foreign capital. Global investors cannot invest in Indian insurers without IFRS-comparable statements. Ind AS 117 brings Indian insurers into the 140+ country IFRS 17 ecosystem.

2. Key Concepts

2.1 The Three Building Blocks of Ind AS 117

Under the General Measurement Model (GMM), every insurance contract group is measured using three components that together form the Insurance Contract Liability on the balance sheet. The liability cannot be less than zero at the group level.

2.1.1 Best Estimate Liability (BEL)

The BEL is the present value of all future cash flows the insurer expects to pay, minus the present value of all future cash flows it expects to receive. It uses probability-weighted, unbiased estimates — the 'best estimate' — discounted at current market rates (not locked-in rates).

```
BEL = EPV(future claims + future expenses) − EPV(future gross premiums)
```

A negative BEL means future premium inflows exceed future outflows in present value terms — the normal state for a profitably priced portfolio. In this model, BEL = ₹−1.607 crore, confirming the portfolio is non-onerous. A positive BEL means the insurer owes more than it will receive — requiring reserves on the balance sheet.

2.1.2 Risk Adjustment (RA)

The Risk Adjustment compensates the insurer for bearing non-financial risk uncertainty — primarily the uncertainty in the amount and timing of mortality claims. Ind AS 117 does not prescribe the calculation method but requires the confidence interval to be disclosed.

Three methods are used in practice:

- **VaR (75th percentile):** Simulate many scenarios; RA = 75th percentile − mean. Used in this model. Most common in Indian practice.
- **Cost of Capital:** RA = PV of cost of holding regulatory capital against insurance risk over the contract lifetime. Common in European Solvency II frameworks.
- **Conditional Tail Expectation (CTE):** RA = average of all outcomes beyond the chosen percentile. More conservative than VaR for the same confidence level.

```
RA = Percentile(simulated claims PV, 75) − Mean(simulated claims PV)
```

2.1.3 Contractual Service Margin (CSM)

The CSM is the most significant innovation of Ind AS 117. It represents the unearned profit in a group of insurance contracts at inception. It defers day-one profit recognition and releases it to P&L as insurance service is delivered.

```
At inception:  CSM = − FCF
where FCF = BEL + RA + acquisition costs − first-year premium received
```

If CSM < 0, the contract group is onerous. The loss is recognised immediately in P&L (Para 47) and CSM is set to zero. This is the most disruptive element of Ind AS 117 for Indian insurers — losses that were previously smoothed or deferred must now be disclosed on day one.

2.2 Factors That Drive the CSM

The CSM is sensitive to several actuarial assumptions. Understanding these drivers is essential for both pricing and financial reporting:

Factor	Effect on CSM	Direction	Example in this model
Profit margin in gross premium	Primary driver	Higher margin → higher CSM	`8% loading → CSM = ₹2.345 cr`
Mortality vs pricing assumption	Experience vs expected	Better mort. → higher CSM	IALM 2012-14 used for both
Discount rate at inception	Affects EPV of cash flows	Higher rate → higher CSM	RBI G-Sec curve, 10yr = 6.53%
Entry age of policyholders	Affects premium adequacy	Younger → higher CSM	Avg age 40.1 — see Section 3.5
Policy term length	Future premiums to collect	Longer → higher CSM	`20-year term modelled`
Lapse rates	Future premiums lost	Higher lapse → lower CSM	Zero lapse assumed (see limits)
Expense overrun vs assumed	Increases outflows	Higher exp. → lower CSM	Fixed 15.6% from IRDAI AR

NOTE The CSM can become negative after inception if mortality experience is significantly worse than priced. In that case, the adverse variance is immediately recognised in P&L (the CSM cannot fall below zero mid-contract). This is why quarterly experience monitoring against pricing assumptions is a core Ind AS 117 operational requirement.

2.3 CSM Roll-Forward — In Full Detail

The CSM Roll-Forward is the formal mechanism by which deferred profit flows into the income statement. It is required to be disclosed in full in the insurer's financial statements under Ind AS 117 Para 101. Each year, four movements are applied:

Movement	Formula	Economic meaning
Opening CSM	Carried from prior year	Unearned profit remaining at start of year

Movement	Formula	Economic meaning
+ Interest accretion	Opening CSM × locked-in rate (6.53%)	CSM grows at the discount rate frozen at inception
− CSM released	(Opening + Interest) × CU_t / ΣCU remaining	Profit earned by delivering insurance service this year
= Closing CSM	Opening + Interest − Released	Unearned profit deferred to next year

Coverage units (CU_t) represent the quantum of insurance service delivered in year t. In this model, CU_t = expected number of policyholders alive in year t. As policyholders age and die, CU_t falls, causing each surviving year's release to represent a growing share of remaining coverage.

The locked-in rate is critical: unlike the BEL (which uses current market rates), the CSM accretes at the rate prevailing at contract inception — 6.53% (the 10-yr G-Sec at FY2025). This rate is never updated. Even if market rates rise to 9% or fall to 4%, the CSM continues to accrete at 6.53%. This design insulates the P&L from interest rate volatility — which flows instead through Other Comprehensive Income (OCI) via the BEL remeasurement.

Three phases are visible in our roll-forward:

- **Years 1–4 (CSM rises from ₹2.345 to peak ₹2.391 cr):** Interest accretion exceeds small early releases. Policyholders are aged ~40–44; IALM mortality is very low (qx ~ 0.002). Few deaths mean few coverage units released.
- **Years 5–12 (gentle decline to ₹2.043 cr):** Releases overtake interest as mortality begins to accelerate. Each year's release grows from ₹0.166 to ₹0.250 cr.
- **Years 13–20 (steep decline to ₹0.000):** Policyholders aged 53–75; IALM qx ~ 0.013–0.038. Coverage units collapse rapidly. Releases grow to ₹0.389 cr in year 20. CSM reaches exactly zero — all profit recognised.

3. Process and Methodology

3.1 Data Sources

Data	Source	URL / Reference
Mortality rates (male)	IALM 2012-14 — IAI	actuariesindia.org/mortality-table
Mortality rates (female)	78% of male (LIC L-42 basis)	IRDAI actuarial practice notes
Discount curve	CCIL Tenorwise G-Sec Yields	ccilindia.com/tenorwise-indicative-yields
Historical G-Sec data	RBI DBIE portal	data.rbi.org.in
Expense ratio (15.6%)	IRDAI Annual Report 2024-25	irdai.gov.in/annual-reports
Claim settlement ratio	IRDAI Annual Report 2024-25	97.82% for life claims
Regulatory standard	Ind AS 117 — MCA Gazette Aug 2024	mca.gov.in

3.2 Mortality Assumption — IALM 2012-14

The Indian Assured Lives Mortality (2012-14) Ultimate table is the IRDAI-mandated standard for Indian life insurance valuation, effective 1 April 2019 under Regulation 5(2) of the IRDAI (Assets, Liabilities and Solvency Margin) Regulations 2016. It was constructed by the Mortality and Morbidity Investigation Committee (MMIC) of the Insurance Information Bureau of India (IIB) using data from 24 life insurance companies over 1 April 2012 to 31 March 2014.

The table covers male insured lives medically underwritten at inception, with a limiting age of 115. Ages are defined as age last birthday. The select period is two years — meaning the table represents ‘duration 2 and over’ ultimate rates, excluding the favourable select effect in the first two years post-underwriting.

Age	Male qx (IALM 2012-14)	Female qx (78% adj.)	Deaths per 1,000 (M)	Annual change in qx
25	0.000931	0.000726	0.93	—
30	0.000977	0.000762	0.98	+5.0%
35	0.001202	0.000938	1.20	+23.0%
40	0.001680	0.001310	1.68	+39.8%
45	0.002579	0.002012	2.58	+53.5%
50	0.004436	0.003460	4.44	+72.0%
55	0.007513	0.005860	7.51	+69.3%
60	0.011162	0.008706	11.16	+48.6%

Age	Male qx (IALM 2012-14)	Female qx (78% adj.)	Deaths per 1,000 (M)	Annual change in qx
65	0.015932	0.012427	15.93	+42.7%

The accelerating rate of change in qx with age is a key driver of CSM release patterns. Mortality roughly doubles every 8–10 years in the working-age population, following the Gompertz-Makeham law: $\mu(x) = A + Bc^x$. This exponential acceleration means coverage units in later policy years represent disproportionately large claims exposure, driving larger CSM releases in years 13–20.

Female Mortality Assumption

The IALM 2012-14 published table covers male lives only. A separate female table exists within the full MMIC investigation report but is not publicly released as a standalone standard. The industry convention, codified in LIC's Valuation Basis L-42, is to apply female mortality rates as a percentage of male rates. This model uses 78% — consistent with LIC practice and the IRDAI actuarial practice notes.

The Institute of Actuaries of India announced in February 2026 that a revised IALM table is ready and awaiting IRDAI approval. This revision is expected to include separately graduated female rates and updated mortality reflecting post-COVID improvements. When adopted, it will require recalibration of all BEL and CSM calculations.

LIMIT The 78% female adjustment is a calibrated approximation, not a graduated female table. It introduces model risk: if actual female mortality diverges from this ratio (evidence suggests Indian urban female mortality may be 70–72% of male in some age bands), the BEL and CSM will be mis-stated. This is flagged as a key limitation to be resolved upon publication of the revised IALM.

3.3 Discount Rate — RBI G-Sec Yield Curve and Cubic Spline Interpolation

Under Ind AS 117 Para 36, discount rates must reflect the characteristics of the insurance contract's cash flows and must be consistent with observable current market prices. Indian actuaries use the RBI Government Securities (G-Sec) yield curve as the risk-free discount curve, available from the Clearing Corporation of India (CCIL) on a daily basis.

Nine tenor points are observed from CCIL (1yr through 30yr). However, cash flows in this model require discount rates at half-year intervals (e.g., t = 0.5, 1.5, … 19.5 for mid-year death claims). Cubic spline interpolation is used to produce a smooth, continuous yield curve across all required durations.

What Cubic Spline Interpolation Is

A cubic spline fits a series of degree-3 polynomials between adjacent observed data points, with two constraints: (1) the curve passes exactly through every observed point, and (2) the first and second derivatives are continuous at every join (ensuring smoothness with no kinks).

The alternatives — linear interpolation and single polynomial fit — are both problematic. Linear interpolation creates kinks at each observed tenor, which is financially unrealistic because yield curves are smooth by nature. A single high-degree polynomial across all nine points can oscillate wildly between observations (Runge's phenomenon), especially at the long end.

The cubic spline is the actuarial and financial industry standard for yield curve construction. scipy.interpolate.CubicSpline with the 'natural' boundary condition (zero second derivative at endpoints) is used in this model, preventing distortion at the 1-year and 30-year boundaries.

NOTE A 10 basis point change in the 10-year G-Sec rate moves the BEL by approximately 0.3–0.5% for a typical term portfolio. At the scale of LIC (which manages ₹32 lakh crore in assets), this represents thousands of crores in balance sheet movement. IRDAI has not yet prescribed a specific interpolation method, so insurers must document and justify their approach in the actuarial function report.

Tenor	G-Sec Yield (FY2025)	Discount factor v(t)	Use in this model
1 year	6.40%	0.9399	Short-tail cash flows
5 year	6.65%	0.7235	Mid-term liabilities
10 year	6.53%	0.5272	Benchmark; locked-in CSM accretion rate
15 year	6.75%	0.3782	Long-duration liabilities
20 year	6.85%	0.2704	Long-tail liabilities
30 year	7.00%	0.1314	Ultra-long tail (not directly used in 20yr model)

3.4 Policy Types — Scope, Assumptions, and Limitations

This model is deliberately scoped to a single product type: a 20-year level term assurance policy paying the sum assured on death only, with level annual premiums payable throughout. This is the most common individual life insurance product in the Indian market and the simplest to model under the GMM. However, the Indian market contains a wide variety of product types, each of which affects calculations differently under Ind AS 117.

Policy type	Death benefit	Survival benefit	BEL character	Ind AS 117 note
Term assurance (modelled)	Sum assured on death	None	Typically negative (profitable)	GMM straightforward
Endowment	Sum assured on death	Sum assured at maturity	Often positive (survival benefit is certain outflow)	GMM; survival benefit adds EPV term

Policy type	Death benefit	Survival benefit	BEL character	Ind AS 117 note
Money-back	Sum assured on death	Periodic % of SA at intervals	Positive in most cases	Multiple cash flow timing points required
Whole life	Sum assured on death (any age)	None	Always positive (death is certain)	Projection to age 115 required
ULIP	Higher of fund or guaranteed death benefit	Fund value	May be investment contract under Ind AS 109	Ind AS 117 scope assessment needed
Group term	Sum assured on death	None	Similar to term, aggregate basis	No individual underwriting; different qx basis
Annuity (deferred)	None (pays on survival)	Regular income for life	Longevity risk (opposite of mortality risk)	IIAM annuitant table used, not IALM
Critical illness	Lump sum on CI diagnosis	None	Morbidity risk; different incidence tables	IAI/Munich Re morbidity tables needed

The impact of product type on this model's outputs is significant:

- **BEL sign:** Term assurance produces a negative BEL (profitable) because death benefits are uncertain and gross premiums are collected with certainty while alive. Endowment produces a positive BEL because the maturity benefit is a certain outflow that dominates the liability.
- **CSM at inception:** Endowments typically generate smaller CSMs (or none) because the certain maturity benefit consumes most of the gross premium loading. Whole life generates larger CSMs due to very long coverage periods but requires extremely long-duration discount rates.
- **RA magnitude:** Critical illness and health products carry larger RAs because morbidity experience is more volatile and harder to predict than mortality. The 75th percentile VaR spread is wider.
- **Contract grouping:** Ind AS 117 Para B92 requires annual cohort grouping and profitability separation. A real portfolio would require separate CSM roll-forwards for each product type within each annual cohort — potentially hundreds of groups.

SCOPE

This model's methodology is directly extensible to other product types. For endowments: add EPV(maturity benefit) to outflows. For money-back: add EPV(survival payments) at each cash flow date. For annuities: replace IALM with the Indian Individual Annuitant's Mortality Table 2012-15 and reverse the mortality risk direction. The core discount curve, CSM mechanics, and Monte Carlo RA framework remain unchanged.

3.5 Portfolio Simulation — Age and Gender Assumptions

Age Distribution

This model uses a uniform random distribution of entry ages from 25 to 55. This is a deliberate simplification that ensures coverage of the full working-age range for a term life portfolio. However, the actual Indian market entry age distribution is bell-shaped, not uniform.

IRDAI Annual Report 2024-25 data and industry practice suggest the following realistic distribution for individual term life in India:

Age band	Approximate market share	Reason	Model treatment
18–25	~8%	Low income, low financial awareness; student loans beginning	Included from 25
26–35	~35%	Peak demand: new earners, first home loan, young family formation	Uniform weight given
36–45	~38%	Highest volume: growing financial responsibilities, employer cover gaps	Uniform weight given
46–55	~17%	Moderate: higher income but higher premiums; children in college	Uniform weight given
56+	~2%	Very low: high premiums, health exclusions, near retirement	Excluded (>55 not modelled)

The uniform assumption produces an average entry age of 40.1 years. A realistic distribution would produce an average of approximately 35–37 years. The effect on results is material:

- **Lower average age → lower average mortality (qx ≈ 0.001–0.002 vs 0.002–0.004):** Net premiums fall, but gross premiums also fall, leaving the profit margin structurally similar.
- **Lower average age → more future premium years at low mortality:** The EPV of inflows increases relative to the EPV of outflows, generating a higher CSM at inception.
- **Lower average age → slower initial CSM release:** Policyholders aged 30–37 have even lower mortality in early years, so years 1–8 would show a longer rising phase before the peak.

For a production model, the age distribution should be calibrated to actual new business data from the insurer's own records — which will vary by distribution channel (bancassurance skews younger; agency channel skews 35–45).

Gender Mix — 70% Male / 30% Female

The 70:30 split is sourced from IRDAI Annual Report 2024-25 and reflects three structural factors: Indian term life was traditionally sold as male breadwinner protection (females covered as riders, not standalone plans); female labour force participation remains ~20–25% nationally (World Bank, 2024), limiting independent purchasing power; and agency networks have been predominantly male, producing a male-skewed acquisition pattern. This ratio is gradually shifting as female workforce participation rises in urban IT and services, Bima Sugam removes agent intermediaries,

and IRDAI's 'Insurance for All by 2047' target drives female financial inclusion. The table below shows the actuarial effect of varying the gender mix on portfolio BEL and CSM.

Scenario	Male %	Female %	Approx. BEL effect	Approx. CSM effect
Current model (IRDAI 2024-25)	70%	30%	Baseline (₹−1.607 cr)	Baseline (₹2.345 cr)
Equal split (future scenario)	50%	50%	More negative (~₹−1.75 cr)	Slightly higher (~₹2.40 cr)
Male-heavy (legacy portfolio)	85%	15%	Less negative (~₹−1.50 cr)	Slightly lower (~₹2.28 cr)

Because female mortality is lower (78% of male), a higher female proportion reduces expected claims PV, making the BEL more negative and the CSM larger — most Indian insurers currently charge unisex rates, implicitly cross-subsidising male lives.

3.6 Premium Calculation

Net premiums are calculated by the equivalence principle: the EPV of future net premiums equals the EPV of future death benefits. This produces the pure insurance cost with no loadings.

```
Net premium P = EPV(death benefit) / EPV(premium annuity)

EPV(death benefit) = Σ[t=1 to T]  tpx × qx+t-1 × SA × v(t-0.5)
EPV(annuity-due)   = Σ[t=1 to T]  tpx × v(t-1)
```

Gross premiums incorporate three loadings, derived from the loss ratio formula:

```
Gross Premium = Net Premium / (1 - expense ratio - commission rate - profit
margin)
              = Net Premium / (1 - 0.156 - 0.050 - 0.080)
              = Net Premium / 0.714
```

Loading component	Rate	Source / Justification
Expense ratio	15.6%	IRDAI Annual Report 2024-25 — management expense ratio for life insurers
Commission rate	5.0%	Typical first-year agent commission for Indian term products
Profit margin	8.0%	Assumed commercial loading; creates positive CSM at inception
Total loading	28.6%	Gross premium = 1.401 × net premium on average

3.7 Claim Payment Timing — In Arrears, In Advance, or Mid-Year

Claim payment timing is a material actuarial assumption that directly affects the BEL. Three conventions exist, each producing a different present value for death benefits:

Convention	Discount factor used	Assumption	Effect on EPV(claims)
In advance (start of year)	v(t−1) = 1/(1+r)^(t−1)	Death paid immediately when it occurs at start of year	Highest — no discounting within the year
Mid-year (UDD) — used here	v(t−0.5) = 1/(1+r)^(t−0.5)	Deaths uniformly distributed; average paid mid-year	Intermediate — half-year discount applied
In arrears (end of year)	v(t) = 1/(1+r)^t	All deaths settled at year end	Lowest — full year discount applied

The Uniform Distribution of Deaths (UDD) assumption — that deaths are uniformly spread across each policy year — is the standard actuarial approximation under both IFoA and IAI practice. It produces a mid-year discount factor of v(t−0.5) = 1/(1+r)^(t−0.5), which is used for all death claim EPVs in this model.

The quantitative impact of timing convention on the BEL in this model:

Convention	Half-year discount factor (10yr rate)	EPV claims (approx.)	BEL impact vs mid-year
In advance	v(9.5) = 0.5509	₹13.93 cr (higher)	+₹0.42 cr (BEL less negative)
Mid-year (used)	v(9.5) = 0.5509	₹13.51 cr (baseline)	Baseline
In arrears	v(10.0) = 0.5272	₹13.08 cr (lower)	−0.43 cr (BEL more negative)

At the portfolio scale of major Indian insurers, a half-year timing difference on a ₹100 crore claims base represents approximately ₹3 crore of balance sheet difference. This is material and must be consistently applied and disclosed. Ind AS 117 Para 33 requires cash flows to be estimated on an expected present value basis reflecting the timing and probability of cash flows — the UDD mid-year convention satisfies this requirement.

Premiums are paid in advance (annuity-due) — at the start of each policy year while the policyholder is alive. This is the standard for Indian life insurance annual premium products and is implemented as v(t−1) for the premium annuity EPV.

3.8 Sum Assured Distribution — Log-Normal and Why

The sum assured for each simulated policy is drawn from a log-normal distribution with mean ₹25 lakh and standard deviation parameter (σ) of 0.55, capped at ₹5 lakh (minimum) and ₹100 lakh (maximum).

```
SA ~ LogNormal(μ=log(25), σ=0.55)
Capped: SA = max(5, min(100, SA))  [in lakhs]
```

Log-normal is chosen for the following statistical and practical reasons:

- **Non-negativity:** Sum assured is a strictly positive quantity. Log-normal is bounded at zero, unlike normal distributions which can produce negative values.
- **Right skew:** The distribution of individual insurance coverage amounts is right-skewed in practice: the majority of policies are in the ₹10–30 lakh range, with a long tail extending to ₹1 crore and above for high-net-worth individuals. Log-normal captures this skew naturally.
- **Multiplicative structure:** Coverage decisions tend to be proportional to income (e.g., '20x annual income' rule of thumb). Income itself is log-normally distributed in India, so coverage derived from income inheritance the same distributional form.
- **Industry standard:** Log-normal is the actuarial standard for individual claim size distributions in life and general insurance. It is used in the loss models that underpin CM2 and CS2 exam syllabuses, and is the default in actuarial software including Prophet and MoSes.

The σ = 0.55 parameter produces a realistic spread: approximately 68% of policies fall between ₹14.4 lakh and ₹43.4 lakh, consistent with IRDAI data showing the bulk of Indian individual term policies in the ₹10L–50L range.

3.9 Monte Carlo Risk Adjustment — Full Methodology

The Risk Adjustment (RA) is calculated via stochastic Monte Carlo simulation. This is the methodologically correct approach for a portfolio of individual lives, and the primary advantage of Python over Excel for Ind AS 117 actuarial work.

Why Monte Carlo for this portfolio

A deterministic model can calculate the expected number of deaths per year (E[deaths] = n × qx) but cannot capture the distribution of deaths around that expectation. The RA compensates for that distributional uncertainty — so the methodology must produce a distribution.

For a portfolio of n independent lives, deaths in a given year follow a Binomial(n, qx) distribution. The Central Limit Theorem means that for large n, this approximates normal. However, across a 20-year projection with heterogeneous ages, genders, and sum assureds, the total claims PV distribution is not analytically tractable — Monte Carlo is required.

Simulation algorithm

```
for sim in range(500):
    total_claims_pv = 0
    for each policy i in portfolio:
        tpx = survival_probabilities(age_i, gender_i, term=20)
        for yr in range(1, 21):
            qx    = IALM_qx(age_i + yr - 1, gender_i)
            death = binomial(n=1, p=tpx[yr] * qx)   # Bernoulli draw
            if death == 1:
                total_claims_pv += SA_i * disc(yr - 0.5)
                break   # policy terminates on first death
    sim_claims[sim] = total_claims_pv

RA = percentile(sim_claims, 75) - mean(sim_claims)
```

Key design choices in this algorithm:

- **Bernoulli draw (binomial n=1):** Each policy in each year is an independent Bernoulli trial. Death probability = tpx[yr] × qx — accounting for both the probability of being alive at the start of year yr and the probability of dying within it. This is actuarially correct.
- **Sequential year simulation:** The loop breaks on first death ('break'). This correctly terminates the policy — a policyholder cannot die twice, and no further premiums or claims arise after death.
- **tpx weighting:** Using tpx[yr] × qx rather than just qx ensures the probability of dying in year yr is the unconditional probability from policy inception, not the conditional probability given survival to year yr. This is the correct formulation for the Bernoulli parameter.
- **500 scenarios:** This produces a stable estimate of the 75th percentile. Increasing to 2,000 scenarios reduces the Monte Carlo standard error by a factor of 2 but requires 4x the computation time. For production models, 5,000–10,000 scenarios are typical.

Interpreting the RA result

The 500 simulations produce total portfolio claims PV values ranging from approximately ₹11.5 crore to ₹15.5 crore, forming a near-normal distribution (Central Limit Theorem applies to the aggregate). Key statistics:

Statistic	Value	Interpretation
Mean	₹13.508 crore	Best estimate of total claims PV (goes into BEL)
Standard deviation	~₹0.62 crore	Actuarial uncertainty in claims PV
75th percentile	₹14.505 crore	Claims PV exceeded in only 25% of scenarios
Risk Adjustment	₹ 0.997 crore	Additional reserve for bearing non-financial uncertainty
RA as % of mean	7.4%	Implicit risk loading above best estimate
Implied confidence	75th percentile	Insurer covers adverse outcomes in 3 of 4 scenarios

4. Results

All results are produced by running IndAS117_Model.py (Python 3) using the data sources and methodology described in Section 3. The model is fully reproducible — setting numpy seed 42 produces identical results on every run.

4.1 Portfolio Summary

Metric	Value	Notes
Total policies	1,000	Simulated; seed 42 for reproducibility
Average entry age	40.1 years	Uniform distribution 25–55 (see Section 3.5)
Male proportion	69.3%	IRDAI 2024-25 industry mix
Female proportion	30.7%	78% mortality adjustment applied
Average sum assured	₹27.7 lakh	Log-normal: μ=log(25), σ=0.55; cap ₹5L–₹100L
Average net premium	₹13,040 p.a.	Equivalence principle; individual qx by age/gender
Average gross premium	₹18,264 p.a.	Net / 0.714 (loadings: 15.6% exp + 5% comm + 8% profit)
Policy type	Level term life	Death benefit only; level annual premium; no survival benefit
Policy term	20 years	Standard Indian individual term product
Claim timing	Mid-year (UDD)	v(t−0.5) discount factor applied to death claims
Premium timing	Annuity-due	v(t−1) discount factor; paid at start of year

4.2 Ind AS 117 Balance Sheet at Inception

The three building blocks of the Insurance Contract Liability (ICL) as at the inception date, before any insurance service has been delivered:

Component	Value (INR crore)	Sign	Economic interpretation
Best Estimate Liability (BEL)	−1.607	Negative	Insurer in net inflow position — profitable portfolio
Risk Adjustment (RA)	+0.997	Positive	Reserve for mortality uncertainty at 75th percentile
Contractual Service Margin (CSM)	+2.345	Positive	Unearned profit deferred over 20-year coverage
Total Insurance Contract Liability	+1.735	Positive	Net liability on balance sheet: BEL + RA + CSM

Derivation of the CSM at inception:

```
FCF at inception = BEL + RA + acquisition costs − first-year premium
                 = −1.607 + 0.997 + (total GP × 5%) − total GP
                 = −1.607 + 0.997 − (total GP × 95%)
                 ≈ −2.345 crore (negative FCF = profitable)

CSM = −FCF = +2.345 crore
```

4.3 Monte Carlo Risk Adjustment Results

Statistic	Value (INR crore)	Interpretation
Mean simulated claims PV	₹13.508	Best estimate; this is what drives the BEL
75th percentile	₹14.505	The threshold: 75% of scenarios are below this
Risk Adjustment (p75 − mean)	₹ 0.997	Additional reserve required for non-financial uncertainty
Approx. standard deviation	₹ 0.62	Spread of outcomes across 500 scenarios
RA as % of mean BEL claims	7.4%	Risk loading implicit in the RA
Scenarios simulated	500	Monte Carlo sample size
Confidence level	75th percentile	Insurer is protected in 75 of 100 future scenarios

4.4 CSM Roll-Forward — Full 20-Year Table

Locked-in accretion rate: 6.53% (10-yr RBI G-Sec at inception, per Ind AS 117 Para 44B). All figures in INR crore.

Year	Age band (avg)	Opening	Interest (6.53%)	Released to P&L	Closing	Phase
1	40–41	2.345	0.153	0.131	2.367	Rising
2	41–42	2.367	0.155	0.139	2.383	Rising
3	42–43	2.383	0.156	0.148	2.391	Peak
4	43–44	2.391	0.156	0.157	2.390	Turning
5	44–45	2.390	0.156	0.166	2.380	Declining
6	45–46	2.380	0.155	0.177	2.359	Declining
7	46–47	2.359	0.154	0.187	2.326	Declining
8	47–48	2.326	0.152	0.199	2.279	Declining
9	48–49	2.279	0.149	0.210	2.217	Declining
10	49–50	2.217	0.145	0.223	2.139	Declining
11	50–51	2.139	0.140	0.236	2.043	Declining
12	51–52	2.043	0.133	0.250	1.926	Declining

Year	Age band (avg)	Opening	Interest (6.53%)	Released to P&L	Closing	Phase
13	52–53	1.926	0.126	0.264	1.788	Accelerating
14	53–54	1.788	0.117	0.279	1.625	Accelerating
15	54–55	1.625	0.106	0.295	1.436	Accelerating
16	55–56	1.436	0.094	0.312	1.218	Accelerating
17	56–57	1.218	0.080	0.329	0.969	Accelerating
18	57–58	0.969	0.063	0.347	0.686	Accelerating
19	58–59	0.686	0.045	0.365	0.365	Accelerating
20	59–75	0.365	0.024	0.389	0.000	Exhausted

5. Interpretation

5.1 What the Negative BEL Means

A BEL of ₹−1.607 crore is not a data error — it is the expected and correct result for a profitably priced insurance portfolio. The EPV of future gross premium inflows (₹18,264 per policy per year across 1,000 policies, discounted at the RBI G-Sec curve) exceeds the EPV of future claims and expense outflows. The surplus is the profit the insurer has locked in at contract inception.

Under Indian GAAP, this profit could be partially recognised upfront on premium receipt. Under Ind AS 117, it flows entirely into the CSM and is released over the 20-year coverage period. This is the fundamental accounting shift the standard introduces — from a cash or premium-receipt basis to a service-delivery basis of profit recognition.

5.2 Age Sensitivity — How Entry Age Affects Every Output

Entry age is the most powerful single driver of individual policy BEL and CSM contribution, because it determines: (a) the entire mortality curve faced over the policy term, (b) the net premium level, and (c) the pattern of CSM release.

Entry age	Net premium (20yr, ₹25L SA)	BEL contribution	CSM contribution	Key mortality range (policy years)
25	~₹3,100 p.a.	Most negative (highest profit)	Highest individual CSM	qx 0.00093–0.012 (ages 25–45)
35	~₹5,900 p.a.	Moderately negative	Moderate CSM	qx 0.00120–0.017 (ages 35–55)
40	~₹8,400 p.a.	Less negative	Lower CSM	qx 0.00168–0.024 (ages 40–60)
45	~₹13,800 p.a.	Near zero or positive	Minimal or zero CSM	qx 0.00258–0.032 (ages 45–65)
50	~₹22,000 p.a.	Positive (higher mortality cost)	May be onerous	qx 0.00444–0.044 (ages 50–70)

A 50-year-old male purchasing a 20-year term policy faces mortality rates that are 4–6 times higher than a 30-year-old in early policy years, and 10–20 times higher in later years. The net premium at age 50 is approximately ₹22,000 versus ₹4,000 at age 30 (for the same ₹25L SA). If the gross premium loading remains at 28.6%, the absolute loading amount is larger for older policyholders — but the uncertainty is also proportionally larger, meaning the RA is higher and the CSM margin may be lower or even negative for the oldest entry ages.

This is why Indian term insurance products typically cap entry age at 50–55 and why policies entered after age 45 are often subject to medical underwriting requirements that change the applicable mortality basis from the IALM ultimate rates to select rates.

5.3 Day-One Profit Prevention — The Core Ind AS 117 Insight

The most important insight from this model is what happens to the ₹2.345 crore CSM: it does not appear in the insurer's income statement in year 1. It sits on the liability side of the balance sheet and is released to P&L only as insurance service is delivered — ₹0.131 crore in year 1, rising to ₹0.389 crore in year 20.

This directly prevents the front-loading of profits that was possible under Indian GAAP. Under Ind AS 117, an insurer selling 10,000 such term policies in FY2027 would record ₹23.45 crore of CSM on day one — not as profit but as a liability. The P&L would show only ₹1.31 crore of insurance service revenue from CSM release in the first year.

KEY	The CSM is not a reserve for claims — it is a reserve for unearned profit. The BEL reserves for claims. The RA reserves for uncertainty. All three together constitute the Insurance Contract Liability. This three-way split is the defining architectural change of Ind AS 117 over Indian GAAP.

5.4 The Risk Adjustment and What It Tells Policyholders

The RA of ₹0.997 crore represents 7.4% of the mean claims PV. This is the financial value the insurer places on uncertainty across 1,000 lives. To understand what this means intuitively: across 500 simulated scenarios, the total portfolio claims PV ranged from approximately ₹11.5 crore (very low mortality year) to ₹15.5 crore (very high mortality year). The insurer chooses to reserve at the 75th percentile — meaning in 375 of 500 scenarios, the actual claims are within the combined BEL + RA buffer.

Under Ind AS 117, the RA confidence level must be disclosed in financial statements. This creates comparability: if HDFC Life uses a 75th percentile and SBI Life uses an 80th percentile, analysts can directly compare their risk margins and draw conclusions about their respective risk appetites. This transparency was impossible under Indian GAAP.

5.5 Limitations and Future Extensions

This model makes several simplifying assumptions. Each represents a known source of model risk that a production-grade Ind AS 117 system must address:

- **Lapse rates (zero assumed):** Indian term products experience lapse rates of 15–20% in policy years 1–3 (IRDAI data). Non-zero lapses reduce EPV(premiums) but also reduce EPV(future claims), with a net effect on BEL that depends on the anti-selection pattern of lapsing policyholders. Incorporating realistic lapses typically reduces the CSM by 5–15%.
- **Expense inflation (zero assumed):** Expenses are modelled as a fixed 15.6% of gross premium. In practice, expense ratios change with inflation, business volumes, and technology investment. A unit-cost model with inflation would increase BEL outflows in later years.

- **Stochastic interest rates (deterministic assumed):** The BEL is calculated using the deterministic RBI G-Sec curve frozen at inception. Ind AS 117 requires the BEL to be remeasured at current rates every quarter. Rate movements flow through OCI (not P&L) but affect the economic balance sheet significantly. A stochastic interest rate model would generate an additional source of RA from interest rate uncertainty.
- **Contract grouping (single group assumed):** Ind AS 117 Para B92 requires annual cohort grouping and profitability tiering. A real portfolio would maintain separate CSM roll-forwards for each product type × annual cohort × profitability tier — potentially hundreds of groups for a large insurer.
- **Age distribution (uniform assumed):** A realistic bell-shaped age distribution peaking at 35–38 would reduce the average entry age to approximately 35 years, improving the portfolio's mortality profile and increasing the CSM by an estimated 8–12%.
- **Female mortality (78% approximation):** The revised IALM table (awaiting IRDAI approval, early 2026) will publish separately graduated female rates. Current approximation introduces model risk of ±5–10% on female policy BEL calculations.
- **Single product type (term only):** Adding endowment, ULIP, or group term would require product-specific cash flow models and potentially different measurement approaches (PAA for short-duration products).

6. Conclusion

This project demonstrates a complete, technically rigorous implementation of the Ind AS 117 General Measurement Model for Indian life insurance. Using real Indian regulatory data — the IALM 2012-14 mortality table, the RBI G-Sec yield curve (cubic spline interpolated), and IRDAI 2024-25 industry benchmarks — the model produces actuarially sound results across all three building blocks.

The key results are: a Best Estimate Liability of ₹−1.607 crore (confirming a profitable, non-onerous portfolio), a Risk Adjustment of ₹0.997 crore at the 75th percentile VaR (derived from 500 Monte Carlo scenarios using np.random.binomial() per policy per year), and a Contractual Service Margin of ₹2.345 crore (the deferred profit that releases to P&L over 20 years, reaching zero at maturity).

This version of the report has significantly enhanced the technical treatment across all dimensions: the three-phase CSM roll-forward is explained in terms of the underlying IALM mortality acceleration; claim timing conventions (mid-year UDD vs in-arrears vs in-advance) are quantified and their balance sheet impact estimated; age sensitivity is documented showing how older entry ages erode the CSM; the gender mix assumption is traced to IRDAI data and its future trajectory analysed; the log-normal sum assured distribution is justified statistically and by market data; the Monte Carlo algorithm is documented at the code level; and the scope limitation to term assurance is explicitly framed with a framework for extending to endowments, whole life, ULIPs, and annuities.

With IRDAI mandating full Ind AS 117 implementation by April 2027, and with the first proforma submission cycle already completed (December 2025), Indian actuaries are now building exactly this kind of model at scale. This project provides a demonstrable, reproducible foundation of that capability. The full Python code, output charts, and this report are available on GitHub.

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